

Hamed Abnavi

Semiconductor Device Engineer – Device Physics

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[LinkedIn](#) — [Google Scholar](#) — [ResearchGate](#)

Profile

PhD researcher with 8+ years of experience in semiconductor devices at the University of Twente, specializing in MOSFET design, TCAD simulation, and power switch architectures across Si, SiC, and GaN technologies. Experienced in the device development cycle: physics-based design, process and circuit simulation, and electrical characterization.

Career Objective: Seeking R&D Engineer, Research Scientist, or Postdoctoral Researcher roles in semiconductor devices and power electronics.

Technical Skills

Device & Process Simulation	Silvaco VictoryDevice & VictoryProcess, Sentaurus TCAD
Circuit Simulation	PSpice, Silvaco Mixedmode, Cadence Virtuoso
Modelling & Analysis	MATLAB, parameter extraction, device modeling, characterisation
Technologies	Si, SiC, GaN, Perovskite; Trench MOSFET, HEMT, Diode, Nanosheet FET
Research Methods	Device engineering, RESURF design, $R_{on,sp} \cdot BV$ trade-off, electrical measurement and characterisation

Professional Experience

PhD Researcher – Power Semiconductor Devices 2023–Present (*expected: 28th Feb. 2027*)
University of Twente, The Netherlands

- Designed and optimized power semiconductor device architectures (trench MOSFETs, bidirectional switches) using physics-based TCAD simulation in Silvaco VictoryDevice and Sentaurus.
- Developed a novel RESURF accumulation-mode trench MOSFET concept as a bidirectional switch, resulting in a first-author publication in *IEEE Transactions on Electron Devices* (2025).
- Characterized fabricated Si devices (transfer/output characteristics, breakdown, parameter extraction) and benchmarked against industrial reference devices from NXP Semiconductors.
- Supervised MSc thesis work on simulation and modeling of GaN bidirectional HEMT devices.
- Conducting ongoing research on nanosheet FET architectures and electrostatic scaling limits for advanced technology nodes.

MSc Researcher – Micro- and Nanoelectronic Devices 2018–2021
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

- Designed, simulated, fabricated, and optimized perovskite solar cells and photodetectors.
- Simulation study on electron/hole transport layers published in *Optical Materials* (2021, 73 citations).

Education

PhD – Power Semiconductor Devices University of Twente, The Netherlands	2023–Present
MSc – Micro- and Nanoelectronics Devices Amirkabir University of Technology (Tehran Polytechnic), Iran	2018–2021
BSc – Electrical Engineering, Control Systems and Robotic	2013–2018

Languages

English (fluent),
Dutch (A2, in progress)

Selected Publications & Awards

Journal Articles

H. Abnavi, C. Steenge, J.W. Berenschot, N.R. Tas, R.J.E. Huetting, “A New Silicon Accumulation-Mode Trench Bidirectional Switch,” *IEEE Transactions on Electron Devices*, 2025. (*Citations: 1*)

H. Abnavi, D.K. Maram, A. Abnavi, “Performance Analysis of Several Electron/Hole Transport Layers in Thin Film MAPbI₃-Based Perovskite Solar Cells: A Simulation Study,” *Optical Materials*, vol. 118, 111258, 2021. (*Citations: 76*)

Conference Papers

H. Abnavi et al., “Accumulation-Mode SiC Trench MOSFET,” *Oral presentation, 17th International Seminar on Power Semiconductors (ISPS)*, 2025.

H. Abnavi et al., “Numerical Approach on Modeling of Perovskite Solar Cells Based on Coupled Ion Vacancy and Charge Carrier Dynamics,” *32nd ICEE*, 2024.

Awards

MESA+ poster award 2025 – 3rd prize – Poster presentation: “Self-aligned 3D transistor technology in the MESA+ NanoLab” at MESA+ day 2025.

References

Available upon request.